Drone to Satellite: Scaling Crop Disease Detection

A step-by-step methodology for taking UAV-based disease detection up to satellite scale, with each step tied to published research

Core idea. A drone sees disease at the scale of single plants but covers only a few fields. A satellite covers whole districts but each pixel mixes many plants. The drone acts as a **bridge**: ground observations calibrate the drone, and drone maps then provide thousands of reliable training labels for the satellite. A review of 2014–2019 UAV–satellite studies found that this kind of **model calibration** is the most common strong synergy between the two platforms (Alvarez-Vanhard et al., 2021). In a wheat study, calibrating Sentinel-2 through UAV data gave R^2 = 0.88, compared with 0.29 when calibrating directly from ground plots (Revill et al., 2020).

Contents

Section	What it covers
1. Why scale from drone to satellite	Strengths and limits of each platform
2. Multi-scale framework	The four levels: spectroscopy, UAV, satellite, forewarning
3. Part A: Drone methodology (A1–A9)	From experimental design to an accurate field disease map
4. Part B: Transfer to satellite (B1–B10)	Spectral, radiometric and spatial harmonization, two-stage training, validation
5. Worked examples	Wheat yellow rust pipeline; rice bacterial leaf blight (Das 2015); Ethiopia rust (Blasch 2023); AVIRIS-NG (Nigam 2019); mustard aphid
6. Challenges and solutions	Mixed pixels, early detection, clouds, small Indian fields
7. Summary of papers	Platform, method, key result and role of each study
8. Interview questions	Short model answers
9. References	Full citations with DOIs

Numbers attributed to a paper come from its published abstract. Numbers marked *illustrative* are examples chosen to explain a calculation and are not results from any study.

1. Why scale from drone to satellite

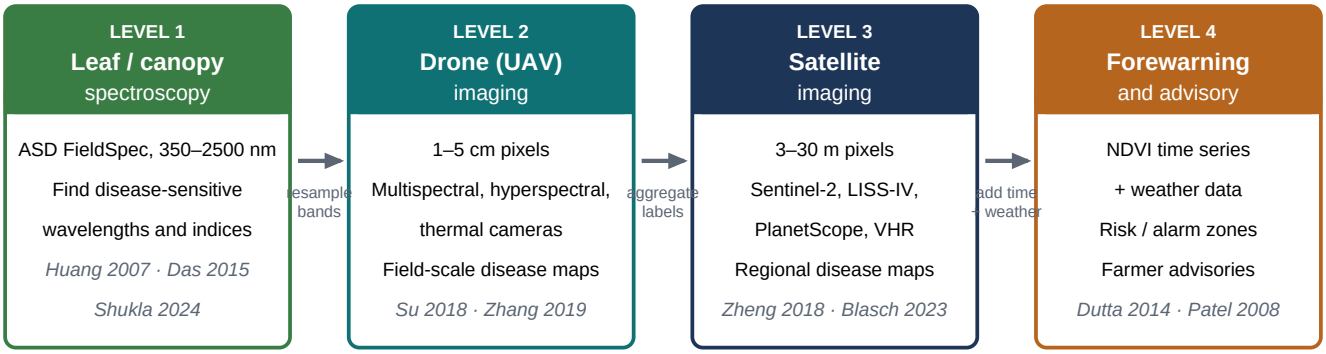
Each platform answers a different question. Drones detect disease early and precisely but cannot cover a state. Satellites cover the state on a regular schedule but see a blend of healthy and diseased plants in each pixel. Combining them gives both accuracy and coverage.

Feature	Ground spectroscopy	Drone (UAV)	Satellite
Pixel / footprint	A few cm (point)	1–5 cm	3–30 m (VHR: 0.3–1 m)
Coverage per day	Tens of plots	10–100 ha	Thousands of km^2
Revisit	On demand	On demand	Fixed (e.g. Sentinel-2: 5 days)
Clouds	Not affected	Flies under cloud	Optical blocked by cloud
Cost per hectare	Very high	High	Low or free (Sentinel-2, Landsat)
Detects	Spectral signature of disease	Individual plants, early patches	Patches large enough to change a pixel
Main role	Find sensitive wavelengths	Accurate maps and training labels	Regional monitoring and forewarning

Evidence for complementarity (Dash et al., 2018). In a forest stress experiment, UAV multispectral imagery detected stress down to individual trees, while RapidEye satellite imagery detected it only in clusters of four or more trees. Resampling the UAV data to satellite resolution showed the gap was not caused by pixel size alone, and each platform needed vegetation indices suited to its own sensor. The authors concluded that UAV data are more sensitive, while satellite data remain a cost-effective way to follow trends over large areas.

2. The multi-scale framework

Disease detection is built up in four levels. Each level discovers or calibrates something that the next level reuses.



Ground truth (disease severity scores, GPS plots) is collected at every level and used to train and validate the next level

Level	Question answered	Typical output	Key papers
1. Spectroscopy	Which wavelengths change when the crop is diseased?	Sensitive bands, indices (e.g. PRI, PSRI)	Huang et al. 2007; Das et al. 2015; Shukla et al. 2024
2. UAV	Where in the field is the disease, and how severe?	Field severity map at cm scale	Su et al. 2018; Zhang et al. 2019; Abdulridha et al. 2019
3. Satellite	How much area is affected across the district?	Regional severity map at 3–30 m	Zheng et al. 2018; Das et al. 2015; Blasch et al. 2023
4. Forewarning	Where will disease appear next?	Risk zones from time series and weather	Dutta et al. 2014; Patel et al. 2008

3. Part A: Drone-based disease detection (Steps A1–A9)

These steps produce an accurate field-scale disease map. Every later satellite step depends on the quality of this map, so the drone stage must be done carefully.

A1	A2	A3	A4	A5	A6	A7	A8	A9
Define symptoms	Design experiment	Ground truth	Choose sensor	Plan flights	Pre-process	Extract features	Build model	Validate and map

A1

Define the disease and its spectral symptoms

Aim: Understand how the disease changes the plant, so the right sensor and wavelengths are chosen.

Procedure

- List the physiological changes: chlorophyll loss (yellowing), pigment changes, leaf structure damage, water loss, rising leaf temperature.
- Map each change to a spectral region: chlorophyll to red and red edge; structure to NIR; water to SWIR; temperature to thermal IR.
- Note the timing: some changes (e.g. stomatal closure, temperature) appear before visible symptoms.

Research basis

- Mahlein (2016) reviews RGB, multispectral, hyperspectral, thermal and chlorophyll fluorescence sensors as objective, reproducible tools for disease detection, and distinguishes the demands of precision agriculture from those of plant phenotyping.
- Zhang J. et al. (2019) review sensors (VIS-NIR, fluorescence, thermal, SAR, LiDAR), feature extraction and algorithms that link remote sensing features to disease and pest occurrence, and propose a general monitoring framework.

Output: A list of expected spectral symptoms and candidate wavelength regions for the target disease.

Disease effect	Spectral region	Example features
Chlorophyll loss, yellowing	Green, red, red edge (500–750 nm)	NDVI, red-edge position, PSSRb, Sentinel-2 B5–B7
Pigment change, senescence	Visible (450–700 nm)	PSRI, PRI (531/570 nm)
Cell structure damage, canopy loss	NIR (750–1300 nm)	NIR reflectance, RVI, OSAVI
Water stress from infection	SWIR (1300–2500 nm)	NDWI / LSWI, water absorption depth
Stomatal closure, transpiration drop	Thermal IR (8–14 µm)	Canopy temperature, CWSI
Visible lesions, pustules	RGB at very high resolution	Colour and texture features, deep learning

A2 Design the experiment

Aim: Create known disease levels so that models can be trained and tested objectively.

Procedure

- Use plots inoculated at graded levels (e.g. healthy, low, medium, high), with replications and untreated healthy controls.
- Include resistant and susceptible varieties, and plots with and without fungicide, to create contrast.
- Add farmers' fields with natural infection for real-world testing.
- Plan repeated observations across the season, because detectability changes with disease development.

Research basis

- Su et al. (2018) used wheat plots inoculated with yellow rust at different levels and flew repeatedly after inoculation.
- Zhang X. et al. (2019) used a controlled field experiment with healthy and rust-infected plots, imaged on five dates across the crop cycle.
- Blasch et al. (2023) used varieties with differing resistance, with and without fungicide, at Kulumsa Agricultural Research Center, Ethiopia.
- Shukla et al. (2024) sampled healthy and aphid-infested mustard in farmers' fields across Bharatpur district, Rajasthan, over two seasons.

Output: A field layout with known disease levels and a measurement calendar.

A3 Collect ground truth and field spectra

Aim: Obtain reliable disease labels and identify the most sensitive wavelengths before flying.

Procedure

- Score disease severity in GPS-tagged quadrats using a standard scale (e.g. percentage leaf area infected, 0–9 scale, or a disease index).
- A common disease index is DI = Σ(number of plants in class × class value) ÷ (total plants × highest class) × 100.
- Measure canopy reflectance with a field spectroradiometer (e.g. ASD FieldSpec, 350–2500 nm) over the same quadrats, against a white reference panel.
- Time ground scoring within a day or two of each flight.

Research basis

- Huang et al. (2007) measured canopy spectra and disease index five times through the season on three wheat varieties and showed that the photochemical reflectance index (PRI) can quantify yellow rust, with potential for airborne or spaceborne sensors.
- Das et al. (2015) found four wavebands (760, 990, 680 and 540 nm) that significantly separated rice infected with bacterial leaf blight from healthy rice.
- Shukla et al. (2024) used ASD FieldSpec 4 spectra; partial least squares regression (PLSR) estimated aphid infestation grade better than principal component regression.

Output: Labelled plots plus a short list of disease-sensitive wavelengths and indices.

A4 Choose the sensor and platform

Aim: Match the camera to the spectral symptoms found in A1 and A3, within budget.

Procedure

- RGB camera: cheap; good for visible lesions and deep learning on texture.
- Multispectral camera (5–10 bands including red edge and NIR): the usual choice; bands can be chosen to match Sentinel-2.
- Hyperspectral camera (hundreds of narrow bands): highest diagnostic power, highest cost and data volume.
- Thermal camera: detects early transpiration changes.
- Platform: multirotor for small plots and low altitude; fixed-wing for larger areas.

Research basis

- Su et al. (2018) used a low-cost five-band multispectral camera flown at 16–24 m, giving 1–1.5 cm pixels.
- Zhang X. et al. (2019) and Abdulridha et al. (2019) used UAV hyperspectral imagers (Abdulridha: 400–1000 nm).
- Revill et al. (2020) used a UAV multispectral sensor whose bands match Sentinel-2, which simplifies the later transfer to satellite.

Output: Sensor and platform choice, ideally with bands that overlap the target satellite's bands.

A5 Plan and fly the missions

Aim: Collect imagery that is geometrically and radiometrically consistent between dates.

Procedure

- Set flight altitude for the required ground sampling distance (GSD); higher altitude covers more area with coarser pixels.
- Use 75–80% forward and side overlap for photogrammetric mosaicking.
- Fly within about two hours of solar noon under stable sky (all clear or uniformly overcast) to limit shadow and BRDF effects.
- Photograph a calibrated reflectance panel before and after each flight; use the camera's downwelling light sensor if available.
- Place ground control points (GCPs) for accurate georeferencing; repeat flights at key growth stages.

Research basis

- These are standard UAV survey practices. Su et al. (2018) flew at low altitude (16–24 m) for centimetre detail; Zhang X. et al. (2019) repeated flights on five dates to follow disease development.

Output: Raw images, panel images, GCP coordinates and flight logs for each date.

A6 Preprocess the imagery

Aim: Convert raw images into an accurate, georeferenced reflectance map.

Procedure

- Radiometric calibration: convert digital numbers to reflectance using the panel (empirical line method) and correct vignetting.
- Build an orthomosaic with structure-from-motion software; georeference with GCPs (target error below 1–2 pixels).
- Mask soil, shadow and weeds (e.g. NDVI threshold or classification) so that only crop canopy is analysed.
- Extract plot or quadrat polygons to link image values with ground scores.

Research basis

- Standard photogrammetric workflow. Kattenborn et al. (2019) also derived 3D predictors from UAV photogrammetry, showing that the same processing yields structural as well as spectral information.

Output: Calibrated reflectance orthomosaic and crop mask for each flight date.

A7 Extract disease features

Aim: Turn reflectance into variables that best separate healthy from diseased crop.

Procedure

- Vegetation indices: NDVI, RVI, OSAVI, NDRE, PRI, PSRI, PSSRb, red-edge position.
- Spectral transforms: first-derivative spectra, continuum removal and absorption band depth.
- Texture features (e.g. grey-level co-occurrence matrix) for patchy diseases; thermal statistics if available.
- Rank features by importance (e.g. random forest importance) and keep the most informative and least correlated.

Research basis

- Su et al. (2018): RVI, NDVI and OSAVI were the best indices, and NIR and red the best bands, for separating healthy from rust-infected wheat.
- Das et al. (2015): of 12 narrow-band indices, PSRI, PSSRb and red-edge position were effective for rice bacterial leaf blight.
- Nigam et al. (2019): wheat yellow rust was identified at field scale using absorption band depth analysis.
- Blasch et al. (2023): the red band and UAV-derived RVI and NDVI were the top predictors of rust progression and yield.

Output: A ranked set of features for modelling.

A8 Build the detection or severity model

Aim: Classify each pixel or plot as healthy or diseased, or estimate severity.

Procedure

- Simple approach: index threshold chosen from the healthy and diseased distributions.
- Machine learning: random forest, support vector machine, k-nearest neighbour, trained on labelled plots.
- Deep learning: convolutional neural networks using spatial and spectral information together.
- Regression (e.g. PLSR) when the target is a continuous severity score.

Research basis

- Su et al. (2018): random forest reached 89.3% accuracy (precision 89.2%, recall 89.4%) at 45 days after inoculation.
- Zhang X. et al. (2019): a deep CNN with Inception-ResNet layers reached overall accuracy 0.85, against 0.77 for a spectral-only random forest.
- Abdulridha et al. (2019): radial basis function and k-nearest neighbour classifiers separated healthy and canker-infected citrus trees with 100% accuracy.
- Shukla et al. (2024): PLSR estimated aphid severity grade from hyperspectral data.

Output: A trained model that produces a field disease map.

A9 Validate and produce the field map

Aim: Prove the map is accurate before using it to train a satellite model.

Procedure

- Hold back independent plots (not used in training); use spatially separated plots to avoid optimistic accuracy.
- Report the confusion matrix, overall accuracy, kappa, precision and recall for classes; R^2 and RMSE for severity.
- Apply the model to whole fields and inspect the map against field notes.
- Record the minimum disease level and stage at which detection becomes reliable.

Research basis

- Su et al. (2018) applied the trained system to a whole farm field. Zhang X. et al. (2019) reported performance was strongest at later stages of disease spread, which shows why the detection stage must be documented.

Output: A validated, georeferenced disease severity map for each flight date.

4. Part B: Transferring the method to satellite (Steps B1–B10)

The drone map is now treated as high-quality reference data. The steps below make the drone and satellite data comparable in wavelength, radiometry, space and time, then train and validate a satellite model.

B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
Pick satellite	Sync dates	Match bands	Match radiometry	Aggregate to pixels	Test features	Train model	Add time + weather	Validate regionally	Operate and advise

B1

Select the satellite

Aim: Choose a sensor whose bands contain the disease-sensitive features from Part A and whose pixel size suits field sizes.

Procedure

- Check which sensitive bands or indices from A7 the satellite can reproduce (e.g. red edge needs Sentinel-2, not LISS-IV).
- Check pixel size against field size: small Indian fields often need 3–6 m pixels or very high resolution.
- Check revisit and cloud risk for the disease season; plan a backup source.

Research basis

- Zheng et al. (2018) chose Sentinel-2 because its red-edge bands B5 and B7 together with red B4 were the most sensitive to yellow rust.
- Das et al. (2015) used LISS-IV (5.8 m) to map rice bacterial leaf blight at village level.
- Blasch et al. (2023) used very high resolution SkySat and Pléiades imagery to follow wheat rust.
- Nigam et al. (2019) found crop classification accuracy of 77–94% with airborne AVIRIS-NG hyperspectral data, against 42–55% with LISS-IV.

Output: Chosen satellite(s) and a list of usable bands and indices.

Sensor	Pixel	Revisit	Disease-relevant bands	Notes
Sentinel-2 MSI	10–20 m	5 days	3 red-edge bands, NIR, 2 SWIR	Free; best general choice; used by Zheng et al. 2018
Resourcesat LISS-IV	5.8 m	5 days (steerable)	Green, red, NIR	No red edge or SWIR; used by Das et al. 2015
Resourcesat LISS-III	23.5 m	24 days	Green, red, NIR, SWIR	Coarser; includes SWIR
PlanetScope	3 m	Daily	Up to 8 bands incl. red edge (newer satellites)	Commercial; frequent revisit helps with cloud
SkySat / Pléiades	0.5–2 m	Tasked	Blue, green, red, NIR	Very high resolution; used by Blasch et al. 2023
PRISMA / EnMAP	30 m	Tasked	Hyperspectral 400–2500 nm	Narrow-band indices from space
AVIRIS-NG (airborne)	4–8 m	Campaign	Hyperspectral 380–2510 nm	ISRO–NASA campaign; used by Nigam et al. 2019
EOS-04 / Sentinel-1 SAR	5–20 m	6–12 days	C-band backscatter	Works through cloud; indirect disease signal (canopy structure)

B2 Synchronize UAV, ground and satellite acquisitions

Aim: Make sure all three observe the crop in the same condition.

Procedure

- Fly the UAV within about ±1–3 days of a cloud-free satellite overpass at the same growth stage.
- Score ground plots on the same dates.
- Repeat at several stages, because UAV–satellite agreement changes as the canopy develops.

Research basis

- Revill et al. (2020) found that agreement between Sentinel-2 and UAV red-edge index values in wheat rose from R^2 = 0.32 at stem elongation to 0.75 at milk development, so the growth stage of matched observations matters.

Output: Matched triplets of satellite image, UAV map and ground scores for each date.

B3 Spectral harmonization: match the bands

Aim: Express UAV or ground spectra exactly as the satellite would see them.

Procedure

- For hyperspectral UAV or field spectra, convolve with the satellite's spectral response function (SRF) for each band:
- $R_{band} = \int R(\lambda) \cdot SRF(\lambda)\, d\lambda \div \int SRF(\lambda)\, d\lambda$
- Recompute indices from the simulated satellite bands and check that they still separate healthy from diseased crop.
- For multispectral UAV cameras, prefer bands that overlap the satellite's bands; otherwise fit a band-to-band regression.

Research basis

- Zheng et al. (2018) converted in situ canopy hyperspectral data to Sentinel-2 bands with the sensor's spectral response function, then designed the Red Edge Disease Stress Index (REDSI) from B4, B5 and B7.
- Das et al. (2015) applied indices derived from ground hyperspectral data to LISS-IV bands to map PSRI and PSSRb.

Output: Disease indices defined in satellite band space and tested on simulated satellite data.

B4 Radiometric harmonization

Aim: Make UAV and satellite reflectance numerically comparable.

Procedure

- Atmospherically correct the satellite image to surface reflectance (e.g. Sen2Cor for Sentinel-2).
- Apply BRDF normalization where view or sun angles differ strongly between dates.
- Compare aggregated UAV reflectance with satellite reflectance over uniform areas; fit a linear correction if there is a bias.
- Mask clouds, cloud shadows and field edges.

Research basis

- Dash et al. (2018) found that UAV and satellite sensitivity differed even after resampling to the same resolution and that each platform needed sensor-appropriate indices, showing that radiometric differences must be handled explicitly.
- Revill et al. (2020) quantified UAV–Sentinel-2 agreement of red-edge indices before using the UAV as a calibration step.

Output: Satellite surface reflectance consistent with UAV reflectance.

B5

Spatial harmonization: aggregate UAV maps to satellite pixels

Aim: Convert centimetre-scale disease maps into one label per satellite pixel.

Procedure

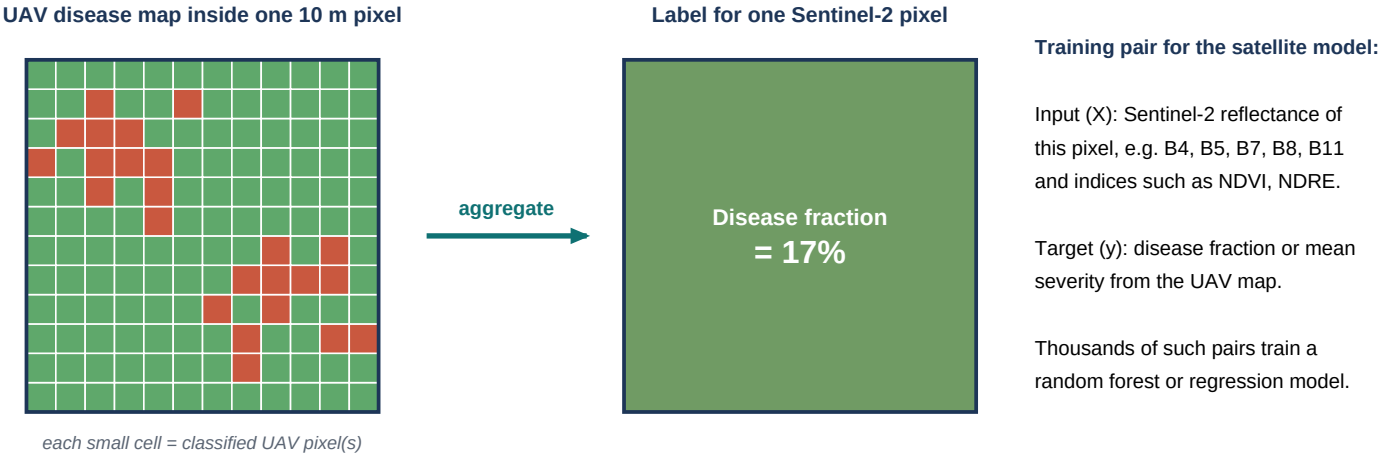
- Co-register the UAV map and satellite image to sub-pixel accuracy (shared GCPs or image matching).
- Overlay the satellite pixel grid on the UAV map.
- For each satellite pixel, compute the disease fraction (infected UAV pixels ÷ all crop UAV pixels) or the mean severity.
- Discard pixels that cross field boundaries, roads or mixed crops; buffer inward from field edges.

Research basis

- Jiang et al. (2023) upscaled UAV-derived wheat maps to the Sentinel-2 grid by pixel aggregation and trained satellite models on the aggregated values.
- Kattenborn et al. (2019) produced cover maps from UAV orthomosaics and used them as reference data for Sentinel-1 and Sentinel-2 models.

Output: A table of satellite pixels, each with a UAV-derived disease fraction or severity label.

Figure: aggregating a UAV disease map to one Sentinel-2 pixel (illustrative)



Illustrative: a UAV flown at about 20 m gives roughly 1.2 cm pixels, so one 10 m × 10 m Sentinel-2 pixel contains about 833 × 833 ≈ 694,000 UAV pixels. The grid above is simplified to 12 × 12 cells: 24 of 144 cells are infected, so the disease fraction is 17%. With real UAV data, if 180,000 of 694,000 crop pixels were infected, the fraction would be about 26%.

B6

Test which features survive at satellite scale

Aim: Find out what the satellite can still detect once healthy and diseased plants are mixed in one pixel.

Procedure

- Plot each candidate satellite index against UAV disease fraction; check the strength and shape of the relationship.
- Estimate the detection limit: the smallest disease fraction that changes the satellite signal beyond normal variation.
- Drop features that worked at UAV scale but not at satellite scale; add satellite-only features (e.g. SWIR, multi-date change).

Research basis

- Dash et al. (2018): satellite data detected stress only in clusters of four or more trees, while UAV data detected single trees.
- Su et al. (2018) found NIR and red most useful at UAV scale, while Zheng et al. (2018) found red plus red-edge bands most useful for Sentinel-2, illustrating that the best features can differ between scales.

Output: Final satellite feature set and a stated detection limit.

B7 Train the satellite model (two-stage calibration)

Aim: Use UAV-derived labels to train a model on satellite data over the whole region.

Procedure

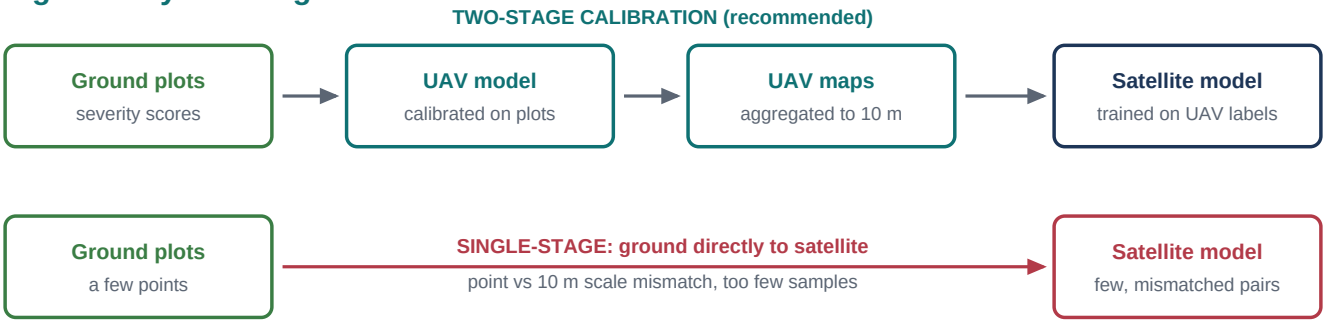
- Inputs (X): satellite bands and indices for each labelled pixel, optionally from several dates.
- Target (y): UAV-derived disease fraction or severity class.
- Algorithms: random forest, Gaussian process regression or regression for severity; random forest or an index threshold for classes.
- Split training and test pixels by field, not at random, to avoid neighbouring pixels leaking into the test set.

Research basis

- Revill et al. (2020): two-stage calibration (ground to UAV to Sentinel-2) gave mean R^2 = 0.88 and NRMSE 8%; single-stage calibration from ground directly to Sentinel-2 gave R^2 = 0.29 and NRMSE 17% (wheat LAI).
- Kattenborn et al. (2019): UAV-based reference data trained random forest models on multitemporal Sentinel-1 and Sentinel-2 data; Sentinel-2 was more important than Sentinel-1, and UAV data were a promising alternative to field surveys when targets are identifiable in UAV imagery.
- Zheng et al. (2018): an optimal threshold on REDSI mapped yellow rust from Sentinel-2 with 84.1% accuracy (kappa 0.76), better than other common disease indices.

Output: A satellite model that maps disease fraction or severity over the region.

Figure: why two-stage calibration works better



Ground plots are small points while satellite pixels are 100 m^2 or more, and there are few plots. The UAV converts a few ground plots into thousands of pixel-sized labels at the right scale.

B8 Add time series and weather for forewarning

Aim: Move from detecting disease that is already present to warning where it is likely to appear.

Procedure

- Build a normal NDVI growth profile for each crop and region from past seasons.
- Flag pixels whose current NDVI falls below normal by more than a threshold at the susceptible stage.
- Combine with weather favourable to the disease or pest (temperature, humidity, leaf wetness) from stations or satellite products.
- Overlay satellite detections to produce risk or alarm zones for advisories.

Research basis

- Dutta et al. (2014) used satellite data to forewarn yellow-rust-affected wheat areas; a later study by Dutta and colleagues combined weather and satellite data to identify stripe rust alarm zones in the 2012–13 and 2013–14 rabi seasons.
- Patel et al. (2008) modelled mustard aphid onset, peak infestation and severity from satellite-derived (TOVS) cumulative air temperature and minimum specific humidity at three sites.

Output: Weekly or per-revisit risk maps that combine current detections with weather-driven risk.

B9

Validate at regional scale

Aim: Show that the satellite map is accurate away from the training fields.

Procedure

- Use independent ground plots and separate UAV flights from fields not used in training.
- Place validation points farther apart than the spatial autocorrelation range so they act as independent checks.
- Report accuracy, kappa, R^2 and RMSE, and compare with agricultural department survey reports where available.
- Test transfer to another district, variety or season before operational use.

Research basis

- Zheng et al. (2018) reported regional accuracy and kappa for Sentinel-2 REDSI mapping. Kattenborn et al. (2019) assessed Sentinel predictions against UAV-based reference maps.

Output: Accuracy statement for the regional disease map, including where it does and does not hold.

B10

Operationalize and disseminate

Aim: Turn maps into timely decisions for farmers and agencies.

Procedure

- Run the satellite model at every cloud-free revisit during the susceptible window.
- Publish severity and risk maps on a geoportal and send village-level advisories (e.g. fungicide timing).
- Fly the UAV in a few sentinel fields each season to update labels and retrain the model (feedback loop).
- Use SAR or higher-revisit imagery when clouds block optical data.

Research basis

- Blasch et al. (2023) presented UAV and satellite rust detection as a first step toward a satellite-enabled early warning system.
- Das et al. (2015) produced village-level LISS-IV maps that distinguished degrees of crop stress.

Output: An operational monitoring and advisory service.

5. Worked examples

5.1 Wheat yellow rust: a complete drone-to-Sentinel-2 pipeline

This example combines methods from several published studies into one pipeline for north-west India. It is a **proposed design**, not a single published study; the research basis for each step is listed.

Step	What is done	Research basis
1. Spectroscopy	Measure canopy spectra and disease index on inoculated wheat plots; confirm sensitivity of PRI and red-edge bands	Huang et al. 2007 (PRI); Zheng et al. 2018
2. UAV design	Plots inoculated at several levels plus healthy controls; resistant and susceptible varieties	Su et al. 2018; Blasch et al. 2023
3. UAV flights	Five-band multispectral camera at about 20 m (about 1–1.5 cm pixels), repeated at tillering, booting, heading	Su et al. 2018
4. UAV model	Random forest on RVI, NDVI, OSAVI and bands (or a CNN if hyperspectral)	Su et al. 2018; Zhang X. et al. 2019
5. Band matching	Simulate Sentinel-2 B4, B5, B7, B8 from field spectra using SRFs; compute REDSI	Zheng et al. 2018
6. Synchronize	Fly within ±2 days of cloud-free Sentinel-2 passes	Revill et al. 2020
7. Aggregate	Disease fraction per 10 m pixel from the UAV map; drop edge pixels	Jiang et al. 2023; Kattenborn et al. 2019
8. Train	Random forest from Sentinel-2 features to UAV disease fraction; compare with REDSI threshold	Revill et al. 2020; Zheng et al. 2018
9. Forewarn	NDVI deviation from normal profile plus favourable weather, giving alarm zones	Dutta et al. 2014; Dutta et al. (stripe rust alarm zones)
10. Validate	Independent fields in another district; accuracy, kappa, RMSE	Zheng et al. 2018

5.2 Rice bacterial leaf blight, Kurnool (Das et al., 2015)

A published example of scaling from ground hyperspectral data directly to satellite, without a UAV stage.

- **Ground spectroscopy:** field hyperspectral observations of healthy and blight-infected rice.
- **Sensitive bands:** 760, 990, 680 and 540 nm significantly separated diseased from healthy crop.
- **Index selection:** of 12 narrow-band indices, PSRI, PSSRb and red-edge position were effective.
- **Satellite transfer:** the index equations were applied to LISS-IV imagery to map PSRI and PSSRb.
- **Result:** village-level maps distinguished different degrees of crop stress.
- **Where a UAV would help:** it would provide many more labelled pixels at LISS-IV scale (5.8 m) than ground plots alone, enabling the two-stage calibration in Step B7.

5.3 Wheat rust in Ethiopia with UAV and very high resolution satellites (Blasch et al., 2023)

- **Platforms:** multispectral UAV, SkySat and Pléiades imagery over a field trial with varieties of differing resistance, with and without fungicide.
- **Features:** 18 spectral features tested as predictors of yellow and stem rust progression and yield loss.
- **Findings:** visible bands were more useful at booting and visible–NIR indices at heading; the red band and UAV-derived RVI and NDVI performed best.
- **Scaling message:** the authors argue the results support upscaling detection from plot to regional level at early growth stages, as a step toward satellite-enabled early warning.

5.4 Airborne hyperspectral bridge: AVIRIS-NG over India (Nigam et al., 2019)

- **Role:** airborne hyperspectral imaging sits between UAV and satellite in scale and previews what spaceborne hyperspectral missions can do.

- **Crop discrimination:** spectral angle mapper and maximum likelihood classification gave 77–94% accuracy with AVIRIS-NG against 42–55% with LISS-IV.
- **Crop health:** LAI and chlorophyll were retrieved with a canopy radiative transfer model; a water condition index separated water stress; wheat yellow rust found at field scale by absorption band depth was then applied to AVIRIS-NG data.

5.5 Mustard aphid: from ground spectroscopy and weather toward satellite monitoring

- **Weather-based forecasting:** Patel et al. (2008) modelled aphid onset, peak and severity from satellite-derived air temperature and humidity.
- **Spectral detection:** Shukla et al. (2024) separated infested from healthy mustard with ground hyperspectral data and estimated severity grade with PLSR.
- **Proposed next step:** fly UAVs over infested and healthy fields in Rajasthan, simulate Sentinel-2 bands from the field spectra (Step B3), aggregate UAV maps to 10 m (Step B5), and combine the resulting satellite detections with the weather-based aphid model (Step B8). This combination has not been published as a single study; it is a research direction.

6. Challenges and solutions

Challenge	Why it happens	Solution
Mixed pixels	A 10 m pixel blends healthy plants, diseased plants, soil and weeds	Predict disease fraction instead of a yes/no class; use higher resolution sensors for small fields
Early disease invisible from space	Early infection affects too few plants to change a pixel (Dash et al. 2018)	Use UAVs for early detection in sentinel fields; use satellites for spread and area
Biotic vs abiotic stress confusion	Nutrient deficiency, drought and disease can all lower NDVI	Use disease-specific bands (red edge, PRI), weather context and multi-date patterns; Nigam et al. 2019 aim to separate biotic and abiotic stress
Clouds during the season	Optical sensors are blocked during rains and fog	Higher-revisit constellations (PlanetScope), compositing, SAR for canopy structure
Time mismatch	Disease develops between UAV flight and satellite pass	Fly within ±1–3 days of overpass; interpolate time series
Radiometric differences	Different sensors, calibration, atmosphere and viewing angles	Atmospheric correction, BRDF normalization, cross-calibration over uniform targets
Small fields in India	Many fields are under 1 ha, so 10–30 m pixels mix several fields	LISS-IV (5.8 m), PlanetScope (3 m) or VHR data; field-boundary masks
Model transfer to new areas	Varieties, soils and disease strains differ	Retrain with a few UAV flights in each new region; test transfer explicitly (Step B9)
Cost of ground truth	Field scoring is slow and subjective	UAV maps as reference data (Kattenborn et al. 2019), with ground plots only to calibrate the UAV

7. Summary of the research papers

Paper	Platform	Target	Method	Key result	Role
Huang et al. 2007	Field + airborne hyperspectral	Wheat yellow rust	PRI vs disease index	PRI can quantify yellow rust	Sensitive index (A3)
Patel et al. 2008	TOVS satellite weather	Mustard aphid	Temperature–humidity model	Predicts onset, peak, severity	Forewarning (B8)
Dutta et al. 2014	Satellite NDVI	Wheat yellow rust	Forewarning of affected areas	Affected areas forewarned	Forewarning (B8)
Das et al. 2015	Ground hyperspectral + LISS-IV	Rice bacterial leaf blight	Narrow-band indices	PSRI, PSSRb, REP effective; village maps	Ground to satellite (5.2)
Mahlein 2016	Review	Plant diseases	Sensor review	Sensors for precision agriculture and phenotyping	Background (A1)
Su et al. 2018	UAV multispectral	Wheat yellow rust	Random forest	89.3% accuracy; RVI, NDVI, OSAVI best	UAV model (A4–A8)
Zheng et al. 2018	Field spectra + Sentinel-2	Wheat yellow rust	REDSI from B4, B5, B7	84.1% accuracy, kappa 0.76	Band matching, satellite model (B3, B7)
Dash et al. 2018	UAV + RapidEye	Forest stress	Time-series comparison	UAV: single trees; satellite: clusters of 4+	Detection limit (B6)
Zhang X. et al. 2019	UAV hyperspectral	Wheat yellow rust	Deep CNN	OA 0.85 vs 0.77 for random forest	UAV model (A8)
Abdulridha et al. 2019	Lab + UAV hyperspectral	Citrus canker	RBF, KNN classifiers	100% tree-level accuracy	UAV model (A8)
Zhang J. et al. 2019	Review	Diseases and pests	Sensors, features, algorithms	General monitoring framework	Background (A1)
Kattenborn et al. 2019	UAV + Sentinel-1/2	Invasive species	UAV labels train random forest	UAV data can replace field sampling	UAV labels (B5, B7)
Nigam et al. 2019	AVIRIS-NG airborne	Crop type and health	SAM, MLC, RTM, band depth	77–94% vs 42–55% (LISS-IV)	Hyperspectral bridge (5.4)
Revill et al. 2020	UAV + Sentinel-2	Wheat LAI	Two-stage GPR calibration	R^2 0.88 vs 0.29 single-stage	Two-stage training (B7)
Alvarez-Vanhard et al. 2021	Review	UAV–satellite synergy	Literature review	Model calibration is the most common strong synergy	Framework (Section 2)
Blasch et al. 2023	UAV + SkySat + Pléiades	Wheat yellow and stem rust	18 spectral features	Red, RVI, NDVI best; supports upscaling	Example (5.3)
Jiang et al. 2023	UAV + Sentinel-2	Wheat growth and N	Pixel aggregation + ML	Satellite models from upscaled UAV maps	Aggregation (B5)
Shukla et al. 2024	Ground hyperspectral	Mustard aphid	PLSR vs PCR	PLSR better for severity grade	Spectroscopy (A3, 5.5)

8. Likely interview questions

Q. Why not detect disease directly from satellite data?

You can for large, developed outbreaks (e.g. Zheng et al. 2018 with Sentinel-2), but early or patchy infection affects too few plants to change a 10 m pixel. UAVs detect it earlier and provide accurate labels to train satellite models.

Q. What is two-stage calibration?

Ground plots calibrate a UAV model; UAV maps aggregated to the satellite grid then calibrate the satellite model. Revill et al. (2020) showed R^2 rising from 0.29 (ground directly to Sentinel-2) to 0.88 (through UAV).

Q. How do you make UAV and satellite data comparable?

Three harmonizations: spectral (convolve spectra with the satellite's spectral response function), radiometric (surface reflectance, BRDF, cross-calibration) and spatial (co-register and aggregate UAV pixels to satellite pixels), plus synchronized dates.

Q. Which Sentinel-2 bands are most useful for wheat yellow rust?

Zheng et al. (2018) found red B4 and red-edge B5 and B7 most sensitive and combined them into REDSI, which mapped rust with 84.1% accuracy.

Q. How would you do this with Indian satellites?

Use LISS-IV (5.8 m) for small fields, as Das et al. (2015) did for rice blight; it lacks red-edge and SWIR bands, so add Sentinel-2 for those. Airborne AVIRIS-NG (Nigam et al. 2019) shows the value of future Indian hyperspectral missions.

Q. How do you move from detection to forewarning?

Combine NDVI deviation from a normal seasonal profile (Dutta et al. 2014) with weather favourable to the pathogen or pest (Patel et al. 2008) to define alarm zones.

9. References

Abdulridha, J., Batuman, O. & Ampatzidis, Y. (2019). UAV-based remote sensing technique to detect citrus canker disease utilizing hyperspectral imaging and machine learning. *Remote Sensing*, 11(11), 1373. https://doi.org/10.3390/rs11111373

Alvarez-Vanhard, E., Corpetti, T. & Houet, T. (2021). UAV & satellite synergies for optical remote sensing applications: A literature review. *Science of Remote Sensing*, 3, 100019. https://doi.org/10.1016/j.srs.2021.100019

Blasch, G., Anberbir, T., Negash, T., Tilahun, L., Belayineh, F.Y., Alemayehu, Y., Mamo, G., Hodson, D.P. & Rodrigues, F.A. (2023). The potential of UAV and very high-resolution satellite imagery for yellow and stem rust detection and phenotyping in Ethiopia. *Scientific Reports*, 13, 16768. https://doi.org/10.1038/s41598-023-43770-y

Das, P.K. et al. (2015). Monitoring of bacterial leaf blight in rice using ground-based hyperspectral and LISS IV satellite data in Kurnool, Andhra Pradesh, India. *International Journal of Pest Management*, 61(4), 359–368.

Dash, J.P., Pearse, G.D. & Watt, M.S. (2018). UAV multispectral imagery can complement satellite data for monitoring forest health. *Remote Sensing*, 10(8), 1216. https://doi.org/10.3390/rs10081216

Dutta, S., Singh, S.K. & Khullar, M. (2014). A case study on forewarning of yellow rust affected areas on wheat crop using satellite data. *Journal of the Indian Society of Remote Sensing*, 42, 335–342. https://doi.org/10.1007/s12524-013-0329-5

Dutta, S., Saharan, M.S. & Sharma, I. (2016). Wheat stripe rust detection of probable alarm zones in Rabi 2012-13 and 2013-14 seasons. *Journal of Wheat Research*. https://epubs.icar.org.in/index.php/JWR/article/view/67185

Huang, W., Lamb, D.W., Niu, Z., Zhang, Y. et al. (2007). Identification of yellow rust in wheat using in-situ spectral reflectance measurements and airborne hyperspectral imaging. *Precision Agriculture*, 8(4–5), 187–197. https://doi.org/10.1007/s11119-007-9038-9

Jiang, J., Atkinson, P.M., Chen, C., Cao, Q., Tian, Y., Zhu, Y., Liu, X. & Cao, W. (2023). Combining UAV and Sentinel-2 satellite multi-spectral images to diagnose crop growth and N status in winter wheat at the county scale. *Field Crops Research*, 294, 108860.

Kattenborn, T., Lopatin, J., Förster, M., Braun, A.C. & Fassnacht, F.E. (2019). UAV data as alternative to field sampling to map woody invasive species based on combined Sentinel-1 and Sentinel-2 data. *Remote Sensing of Environment*, 227, 61–73. https://doi.org/10.1016/j.rse.2019.03.025

Mahlein, A.-K. (2016). Plant disease detection by imaging sensors – parallels and specific demands for precision agriculture and plant phenotyping. *Plant Disease*, 100(2), 241–251. https://doi.org/10.1094/PDIS-03-15-0340-FE

Nigam, R. et al. (2019). Crop type discrimination and health assessment using hyperspectral imaging. *Current Science*, 116(7), 1108 onwards.

Patel, N.K., Chattopadhyay, C., Dutta, S. et al. (2008). Modelling regional level spatial distribution of aphid (*Lipaphis erysimi*) growth in Indian mustard using satellite-based remote sensing data. *International Journal of Pest Management*, 54, 51–62. https://doi.org/10.1080/09670870701472314

Revill, A., Florence, A., MacArthur, A., Hoad, S., Rees, R. & Williams, M. (2020). Quantifying uncertainty and bridging the scaling gap in the retrieval of leaf area index by coupling Sentinel-2 and UAV observations. *Remote Sensing*, 12(11), 1843. https://doi.org/10.3390/rs12111843

Shukla et al. (2024). Detection of aphid-infested mustard crop using ground spectroscopy. *Remote Sensing*, 16(1), 47. https://doi.org/10.3390/rs16010047

Su, J., Liu, C., Coombes, M., Hu, X., Wang, C., Xu, X., Li, Q., Guo, L. & Chen, W.-H. (2018). Wheat yellow rust monitoring by learning from multispectral UAV aerial imagery. *Computers and Electronics in Agriculture*, 155, 157–166. https://doi.org/10.1016/j.compag.2018.10.017

Zhang, J., Huang, Y., Pu, R., Gonzalez-Moreno, P., Yuan, L., Wu, K. & Huang, W. (2019). Monitoring plant diseases and pests through remote sensing technology: A review. *Computers and Electronics in Agriculture*, 165, 104943. https://doi.org/10.1016/j.compag.2019.104943

Zhang, X. et al. (2019). A deep learning-based approach for automated yellow rust disease detection from high-resolution hyperspectral UAV images. *Remote Sensing*, 11(13), 1554. https://doi.org/10.3390/rs11131554

Zheng, Q., Huang, W., Cui, X., Shi, Y. & Liu, L. (2018). New spectral index for detecting wheat yellow rust using Sentinel-2 multispectral imagery. *Sensors*, 18(3), 868. https://doi.org/10.3390/s18030868

Note: author lists given as "et al." were not fully confirmed from accessible sources. Details attributed to papers are taken from their abstracts; consult the full texts before quoting exact figures in writing.